SOMAIYA VIDYAVIHAR UNIVERSITY

<table>
<tr><td colspan="3">Semester: January 2025 – April 2025
Maximum Marks: 30 **Examination: In-Semester Examination** **Duration:** 1hr.15 Min</td></tr>
<tr><td>**Programme code:** 03
Programme: B.Tech Electronics and Telecommunication Engineering</td><td>**Class:** SY</td><td>**Semester:** IV
(SVU 2023)</td></tr>
<tr><td>**Institute/School/ Department:**
K. J. Somaiya School of Engineering</td><td colspan="2">**Name of the department:** EXTC</td></tr>
<tr><td>**Course Code:** 216U03C403</td><td colspan="2">**Name of the Course:** Communication Systems</td></tr>
</table>

<table>
<tr><th>Question No.</th><th>Marking Scheme</th><th>Max. Marks</th></tr>
<tr><td>Q1</td><td>What is significance of Noise Figure in communication system? Certain amplifier in television receiving system has a signal to noise ratio of 20 dB at input, and becomes 15 dB at output after the signal is passing through it. Calculate noise figure in ratio and decibels.
<table>
<tr><td>**Significance of noise figure:** Noise figure value tells how much signal to noise ratio decreases from input to output of any system.</td><td>1 Mark</td></tr>
<tr><td>SNR (dB)=10 log [SNR (ratio)]
Therefore, SNR(ratio)= antilog($\frac{SNR\ (dB)}{10}$)
Input SNR = 100</td><td>1 Mark</td></tr>
<tr><td>SNR (dB)=10 log [SNR (ratio)]
Therefore, SNR(ratio)= antilog($\frac{SNR\ (dB)}{10}$)
Output SNR = 31.62</td><td>1 Mark</td></tr>
<tr><td>Noise Factor=$\frac{Input\ SNR}{Output\ SNR}$
= $\frac{100}{31.62}$ =**3.162**</td><td>1 Mark</td></tr>
<tr><td>Noise Figure = 10 log (Noise Factor)=**4.99 dB≈ 5 dB**</td><td>1 Mark</td></tr>
</table>
<u>OR</u>
What is correlated noise? Certain nonlinear amplifier is given a two input signals of 2 KHz and 12 KHz. Calculate the sum and difference frequencies which are present at the output of this amplifier for values of m and n of 2 and 3.
<table>
<tr><td>Definition of correlated noise</td><td>1 Mark</td></tr>
<tr><td><table>
<tr><th>m</th><th>n</th><th>F1(2KHz)</th><th>F2(12 KHz)</th><th>Sum</th><th>Difference</th></tr>
<tr><td>2</td><td>3</td><td>4 KHz</td><td>36 KHz</td><td>40 KHz</td><td>32 KHz</td></tr>
<tr><td>3</td><td>2</td><td>6 KHz</td><td>24 KHz</td><td>30 KHz</td><td>18 KHz</td></tr>
<tr><td>2</td><td>2</td><td>4 KHz</td><td>24 KHz</td><td>28 KHz</td><td>20 KHz</td></tr>
<tr><td>3</td><td>3</td><td>6 KHz</td><td>36 KHz</td><td>42 KHz</td><td>30 KHz</td></tr>
</table></td><td>1 Mark for each value</td></tr>
</table></td><td>05</td></tr>
<tr><td>Q2</td><td>a. Draw time domain representation of AM wave (on graph paper) having modulating signal of $10\sin(4\pi\times 250t)$ and carrier signal of $25\cos(4\pi\times 10^6 t)$ Calculate modulation index.
<table>
<tr><td>Labeled modulating signal (sine wave)</td><td>1 mark</td></tr>
<tr><td>Labeled carrier signal (cosine wave)</td><td>1 mark</td></tr>
<tr><td>Labeled AM wave</td><td>1 mark</td></tr>
<tr><td>Modulation index = $\frac{V_m}{V_c} = \frac{10}{25} = 0.25$</td><td></td></tr>
</table></td><td>05</td></tr>
</table>

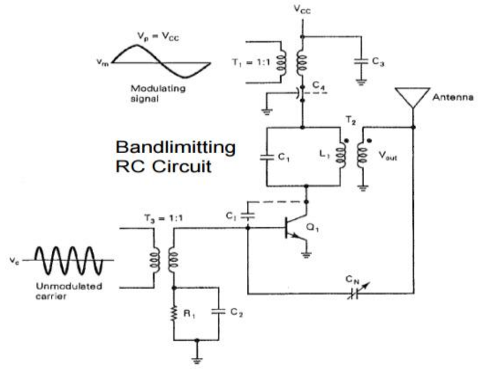
b. Draw the amplitude spectrum for the same signal representing all values and show calculation.

Amplitude spectrum with neat labeling	2 marks

Q3 — 05

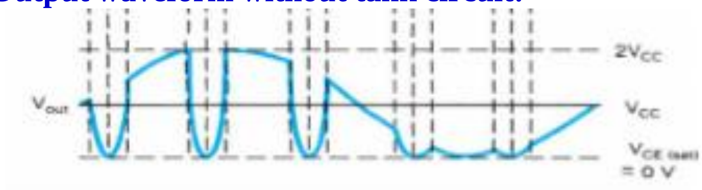
Draw neat labeled circuit of medium power AM modulator. What is role of tank circuit in it? Draw the output waveforms of this circuit with and without tank circuit.

Circuit diagram of medium power AM modulator: — 2 marks

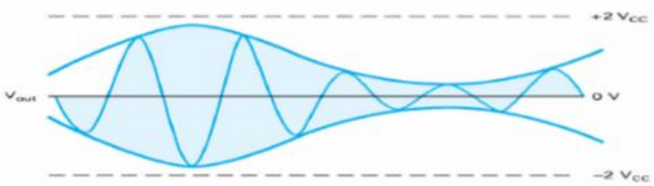


Role of tank circuit: For the flywheel effect and generate negative half cycle of the AM signal — 1 mark

Output waveform without tank circuit: — 2 marks



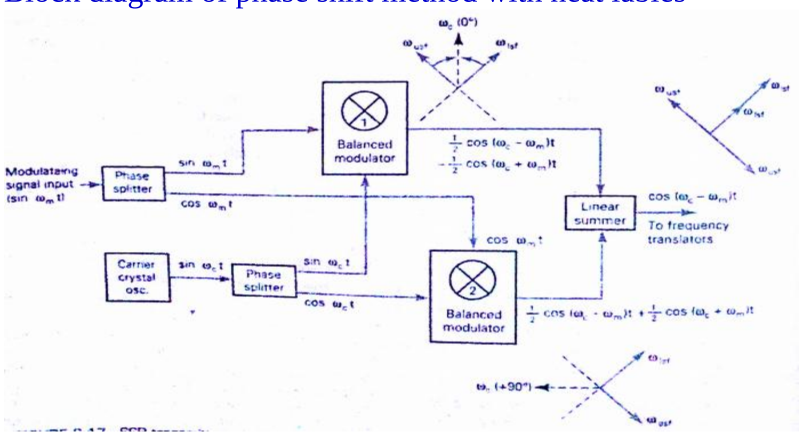
Output waveform with tank circuit:



OR

Draw neat labeled block diagram of phase shift method of single sideband AM generation. Mathematically prove that use of phase shifter removes one sideband from the signal.

Block diagram of phase shift method with neat lables — 3 marks



Mathematical expression — 2 marks

... outputs are expressed mathematically as

output from balanced modulator 1 $= (\sin \omega_m t)(\sin \omega_c t)$

$$= \frac{1}{2}\cos(\omega_c - \omega_m)t - \frac{1}{2}\cos(\omega_c + \omega_m)t$$

output from balanced modulator 2 $= (\cos \omega_m t)(\cos \omega_c t)$

$$= \frac{1}{2}\cos(\omega_c - \omega_m)t + \frac{1}{2}\cos(\omega_c + \omega_m)t$$

and the output from the linear summer is

$$\frac{1}{2}\cos(\omega_c - \omega_m)t - \frac{1}{2}\cos(\omega_c + \omega_m)t$$

$$+\frac{1}{2}\cos(\omega_c - \omega_m)t + \frac{1}{2}\cos(\omega_c + \omega_m)t$$

$$\underbrace{\cos(\omega_c - \omega_m)t}_{\text{lower sideband (difference signal)}} \qquad \text{canceled}$$

Q4

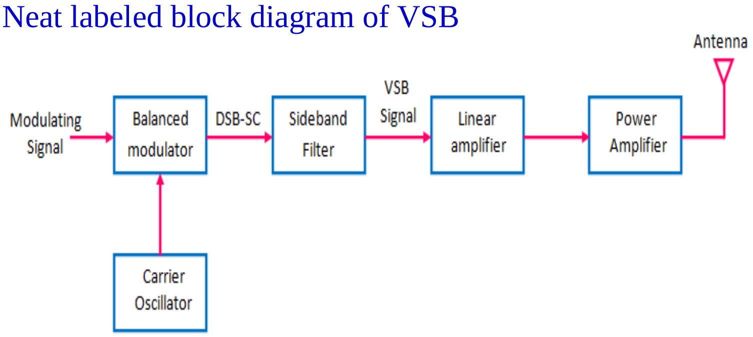
Attempt ALL of the following questions:

A. What is role of RC time constant in demodulation of AM signal? Explain with diagram. — 2 marks

The RC time constant decides the rate of charging and discharging of capacitor used as a filtering element in demodulator circuit of AM. Waveforms of high and low value of RC time constant	2 marks

B. Draw the neat labeled block diagram of VSB signal generation. — 2 marks

Neat labeled block diagram of VSB — 2 marks



C. What is role of channel communication system? Give two practical example of it. — 2 marks

Channel: Refers to the physical medium through which and electrical signal carrying information is transmitted from transmitter to receiver.	1 mark
Example: Air, optical fiber cable, coaxial cables, microwave links, satellite links etc.	1 mark

D. Calculate the carrier power and sideband power of AM wave in a load resistance of 100 Ω if the peak voltage of carrier wave is 100 V and modulation index is 0.4. What is percentage of sideband power? — 2 marks

$Carrier\ Power = \frac{V_c^2}{2R} = \frac{(100)^2}{2\times100}$=50 Watts	1 Mark
Sideband power=$\frac{m^2V_c^2}{4R} = \frac{(0.4)^2\times(100)^2}{4\times100} = 4\ Watts$	1 Mark

E. Write meaning of each term with units in the following equation: — 1 mark

$$V_N = \sqrt{4RKTB}$$

Meaning of each term with units	1 Mark

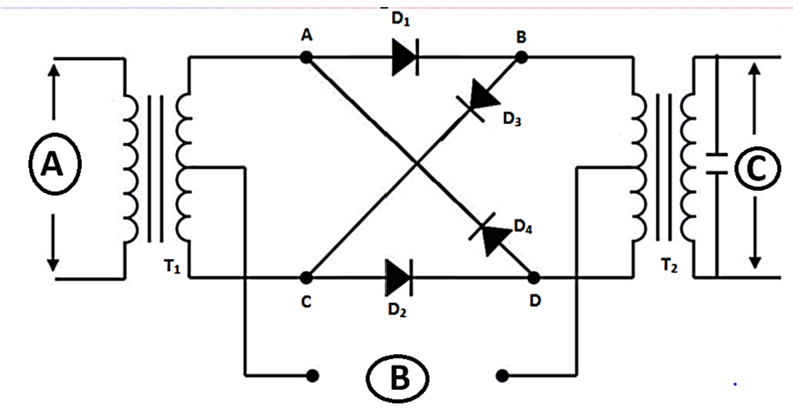
F. Write the odd word from the list given below: — 1 mark

Industrial noise	Thermal noise	Shot noise	Intermodulation noise	Transit time noise

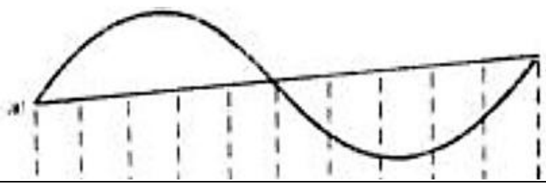
Intermodulation noise is correlated noise, rest of them are uncorrelated noise	1 mark

Q5 — 05

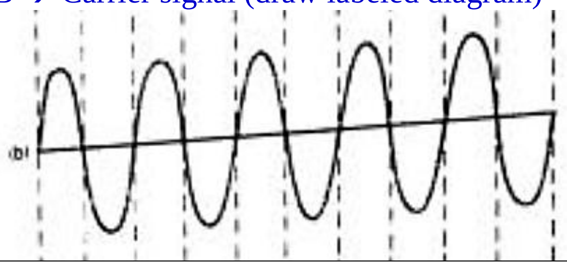
Name the circuit given below. Draw the signals at point A, B and C giving neat labeling. Mention one advantage of using this circuit.



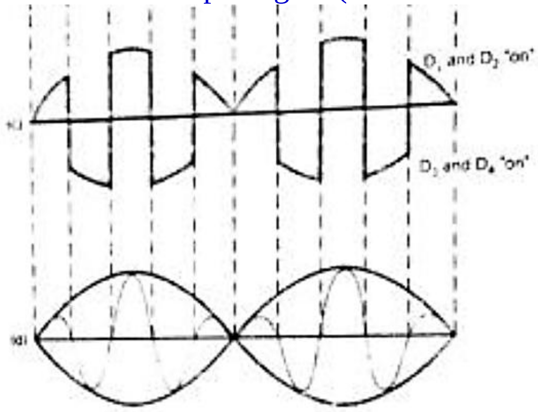
Name of circuit: Ring modulator for DSBSC generation	1 mark
A → Modulating signal (draw labeled diagram)	1 mark



B → Carrier signal (draw labeled diagram)	1 mark



C → DSBSC output signal (draw labeled diagram)	1 mark

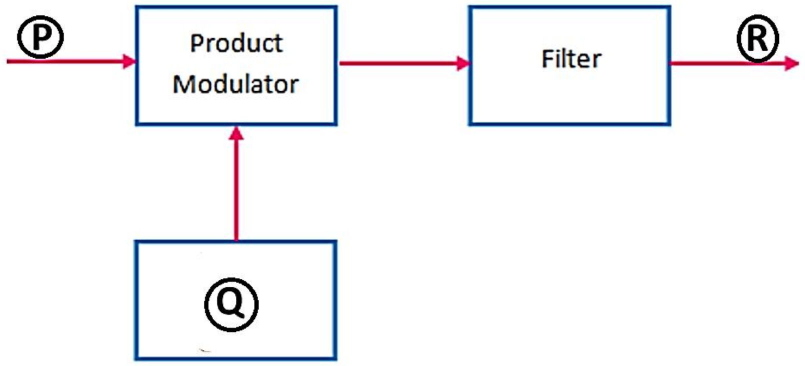


Advantage of circuit: (anyone) — 1 mark

- Inherent ability to effectively suppress carrier signal, producing pure sideband output without using filters
- Effective method of generating DSBSC signal with minimal power consumption

OR

Name the figure given below. Write the name of signals/circuits/elements present at point P, Q and R. With mathematical equations obtain the signal at point R.



Name of circuit: Ring modulator for SSBSC or DSBSC demodulation	1 mark
P → DSBSC or SSBSC	1 mark
Q → Locally generated carrier	1 mark
R → Modulating signal	1 mark
Mathematical expression: Thus, $m(t) = \frac{1}{2} x(t) E_c [\cos(4\pi f_c t + \phi) + \cos \phi]$ Therefore, $m(t) = \underbrace{\frac{1}{2} E_c \cos \phi\, x(t)}_{\text{Scaled version of message signal } x(t)} + \underbrace{\frac{1}{2} x(t) E_c \cos(4\pi f_c t + \phi)}_{\text{Unwanted term}}$ $v_o(t) = \frac{1}{2} E_c \cos \phi\, x(t)$	1 mark